

CODE SPORT



In this month's column, we feature a set of interview questions on algorithms, data structures, operating systems and computer architecture.

For the past few months, we have been discussing information retrieval, natural language processing (NLP) and the algorithms associated with them. In this month's column, we take a break from our discussion on NLP and explore a bunch of computer science interview questions.

1. You are asked to write a small code snippet in C to determine whether your computer is a little endian machine or a big endian machine. Assume that you can compile and run your C code using your favourite compiler on this computer. Is it possible for you to determine the endianness by looking at the assembly code without running the program? If not, explain how you can determine the endianness of the machine by running your code snippet.
2. You are running a C program and from the console you see that your program terminated with a 'stack overflow' signal error. When would a stack overflow signal be generated?
3. We all know that physical memory is volatile and, hence, when there is a power failure or system shutdown, contents of the physical memory are lost. Is it possible to have non-volatile physical memory? What are the different types of non-volatile physical memory?
4. If your computer has non-volatile physical memory instead of volatile physical memory, can you explain what would be the major operating system support needed to enable an application to restart from where it crashed?
5. Why do computer systems have multiple levels of cache hierarchy instead of a single large huge cache? Is it possible for an application to bypass the cache and read/write directly from the main memory?
6. We are all very familiar with Moore's law, which can be approximately stated as: "The transistor count on a processor chip doubles every 18 months." This has been resulting in the approximate doubling of

processor speeds till now. However, over the last five years, there has been considerable talk about the end of Moore's law. What are the factors that are causing the end of Moore's law? Can you explain what the term 'dark silicon' means?

7. What is meant by 'Non-Uniform Memory Access (NUMA)' systems? If you have written a C application that runs on your personal computer, do you need to make changes to it in order for it to run on a NUMA system? If yes, what changes are needed? If not, explain why no changes are needed.
8. Given the wide prevalence of Android mobile devices today, what is the operating system running on these devices? If you have written a Java program that runs on your personal computer, can you run it without any changes on your Android mobile system? If not, explain why it can't be run, as is.
9. We all know that the data present on our laptops and personal computers is typically organised into files and directories. What are the different storage abstractions that are available on Android mobile devices for mobile applications to store data?
10. Many computer systems today have both a CPU and GPU built in to the same system, in what is known as 'Heterogeneous System Architecture (HSA)'. Is it possible for the same memory address space to be shared by the CPU and GPU? If yes, explain how it can be shared. If not, explain how data will be transferred between CPU and GPU.
11. We are familiar with various search algorithms such as Breadth First Search (BFS) and Depth First Search (DFS). If you are asked to pick a specific search algorithm to be run on a machine that is limited by the amount of physical memory available, which search algorithm